

Robotics

chap No 01:-

mechanical manipulator (An arm robot).
Machines which are for the most part
limited to one class of task are
considered fixed automation.
manipulator has six rotational joints.

Description of position & orientation:

In robotics, we often deal with objects placed
in three-dimensional space, such as robot arms,
tools and parts in robotic environment.

1- Position:-

- position tells us where an object is located in 3d space.
- Usually described using co-ordinates (x, y, z) .

2- Orientation:-

- tells us how an object is rotated or aligned in space.
- describes the direction an object is facing.

Co-ordinate frames:-

To describe position & orientation this frame is attached to each object. It moves rigidly with the object. Objects position & orientation are described relative to a reference frame (base or world frame).

Reference frame transformation:

A co-ordinate frame can be used as a reference. Sometimes we need to change or transform the description of an object from one frame to another. These are handled using mathematical methods.

* Proper representation of position & orientation is essential in robotics for motion, control & manipulation.

Kinematics:-

The study of motion without considering forces includes:- position, Velocity, Acceleration, higher-order derivatives.

* In robotics, geometric & motion properties of manipulators.

Structure of manipulator:-

A manipulator is made up of rigid links connected by joints. Joints allow movement between links.

1 - Revolute (Rotatory) joint:- Motion is rotational. (joint angle)

2 - Prismatic (sliding) joint:- Motion is linear (joint offset).

Degrees of freedom:-

Number of independent joint variables needed to describe robot's position. In most industrial robots:- No of joints = No of DOF

e.g:- A mechanism may have many parts but fewer DOF.

End-effector: The device at the end of robot arm. e.g. - Gripper, welding torch, electromagnet.
Its position is described using a tool frame.
The fixed base has a base frame.

Forward kinetics:-

Problem of finding the position & orientation of end effector when the joint angles (or displacement) is given.

Input $\rightarrow$ joint angle (joint space)

Output $\rightarrow$ End-effector position & orientation (Cartesian space)

It converts:-

joint space $\rightarrow$ Cartesian space.

Inverse kinetics:-

Problem of finding the joint angles when the desired position & orientation of the end-effector are given.

Input $\rightarrow$ End-effector position & orientation.

Output $\rightarrow$ joint angles.

It converts:-

Cartesian space $\rightarrow$ joint space.

essential for real world robot control.
When we give a robot a target position in 3D, it must calculate: how much each joint should rotate or slide.
This calculation done using an algorithm in robot's control system.

Inverse kinematics is more complex than forward kinematics because: kinematic equations are non-linear.

A solution may have:-

- Have multiple possible joint configurations.
- Have no solution.
- Be difficult to solve mathematically (no closed-form solution).

By Cartesian space, we mean the space in which the position of a point is given with 3 numbers. It is the orientation of a body is given with 3 numbers. It is sometimes called task space or operational space.

Why important??

Why difficult?

Workspace concept:- workspace is the set of all positions & orientations the robot can reach. if no inverse kinematics solution exists:-

- The target lies outside the workspace.
- The robot cannot reach that position.

Early Vs Modern robots:

- Early used records & playback (joint values were stored manually)
- Modern industrial use inverse kinematics automatically.

Velocity analysis & jacobian:- When a robot is moving we analyse its velocities.

To relate joint motion to the end-effector motion, we use a matrix called the jacobian. It maps:- joint velocities $\rightarrow$ End-effector velocities
joint space $\rightarrow$ cartesian space.

joint rates $\rightarrow$ jacobian $\rightarrow$ End-effector Velocity

The jacobian changes depending on the robot's configuration.

Singularities:- It occurs when:-

- The jacobian matrix becomes non-invertible.
- The robot loses one effective degree of freedom.
- Some motions require very high or infinite joint speed.
e.g. Rear Gunner mounted with:-
 - Azimuth rotation (left-right)
 - Elevation rotation (up-down)

Why singularities matter in Robots??

The robot can still reach the position but controlling velocity near that position becomes difficult. Joint speeds may become extremely large.

All manipulators have singularities - they can't completely avoid.

Static Forces & Jacobian:-

forces instead of moving. e.g.:- Robots sometimes apply forces instead of moving. e.g.:-
• Pressing a surface.
• Holding an object.

Q: What joint torque needed to produce a desired force at the end-effector?

Again, the jacobian matrix is used to relate:
End-effector force $\longleftrightarrow$ joint Torques.

Robotics (Dynamics):

It is the study of the forces & torques required to produce motion.

To move a robot manipulator:-

1. Start from rest 2. Move at constant velocity 3. stop smoothly.

The joint actuators must apply specific torque functions.

→ Torque depends on:-

- path of motion
- speed & acceleration
- Mass of links
- payload weight
- Friction in joints.

Human naturally consider weight when lifting objects. if we misjudge the weight, motion becomes incorrect.

Similarly, robot control algo must consider:-

- Mass (amount of matter in an object) Kilogram
- Inertia (resistance of an object to change its state of motion)
- Dynamic effects (the effects caused by forces during motion)

Not just position (kinematics).

- Uses of Dynamic Equations:
1. control calculate required joint torques to follow a desired path.
 2. simulations predict robot motion when torques are applied.

Trajectory Generation:-

A ~~robotic~~ process of planning how a robot's joints should move over time to reach a desired position smoothly and in a controlled manner.

- Each joint follows a smooth function of time.
- Usually, all joints start and stop together for coordinated motion.

- Via points & splines:-
1. Sometimes the robot must pass through intermediate points.
 2. A spline is a smooth curve that passes through these points.
- Cartesian Trajectory:- if the end-effector must follow a straight line or specific shape in space:
- The motion in Cartesian space must be converted into joint motions.

Design considerations:-

Manipulator design depends on the intended task, economics & physical requirement

key factor:-

- Size, speed, load capacity
- Number of joints & their arrangements.
- workspace size & structural stiffness.

More joints → More dexterity but higher cost & complexity.

Two design approaches:-

1. Specialized robot:- Designed for a specific task (4 joints for PCB assembly)

2-Universal robot: performs many tasks; minimum 6 joints for full flexibility in 3D space.

sensors & actuators:-

- Actuators move the joints (motors, hydraulic/pneumatic systems).
- Transmission systems transfer actuator motion to joints.
- Internal sensors measure joint position; some also measure forces.
- Sensor placement affects accuracy, control & performance.

linear position control:-

Some manipulators move directly with stepper motors. Most require torque computation to follow a desired trajectory. position control system use feedback from position and velocity sensors to correct errors and suppress disturbances. Linear control methods are common in industry.

Non-linear position control:-

Considers full non-linear dynamics of the manipulator. promise better performance than linear. Modern robots use this algo.

Force - control:-

Important when manipulator contacts surfaces or objects. position control alone may cause excessive force. Loss of contact - hybrid control:- some directions controlled by position, others by force.
e.g. window washing: maintain force perpendicular to glass, move along plane tangentially.

Robot programming:

- humans to define robot motion & tasks.
- Movements are described using a tool center point relative to a reference coordinate system.
- paths are specified by points, speeds, motion modifiers.
- Sophisticated programming allows flexible, sensor-based multi-robot coordination.

off-line programming & simulation:

programming development without using actual robots.

Uses computer graphics & CAD integration to

① Reduce programming time

② Keep production running while reprogramming

supports simulation of motion & task verification.

Notation in Robotics:

Uppercase $\rightarrow$ vectors/matrices; lowercase $\rightarrow$ scalars.

Leading subscripts/superscripts $\rightarrow$ coordinate system reference

Trailing superscripts $\rightarrow$ inverse or transpose.

Vectors $\rightarrow$ column vectors; row vector use transpose.

Angular velocities of links sum ~~velocity~~ vectorially

only if expressed in the same frame.